Reg. No. | | | | | | | | | |

Question Paper Code : 14646

B.E. / B.TECH. DEGREE EXAMINATION, NOVEMBER / DECEMBER 2018

Third Semester

Common to B.E. – Computer Science & Engineering and B.Tech. - Information Technology

15MA304 – MATHEMATICS III - DISCRETE MATHEMATICS

(Regulations: Mepco – R2015)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. The number of possible subsets of a set A = $\{a,b,c,d,e\}$ is _____ .
2. Define Poset.
3. Using truth table, show that $p \vee \neg(p \wedge q)$ is a tautology.
4. Negate the statement "Every student in this class is intelligent."
5. A graph which contains no more than one edge between a pair of vertex is

   A) undirected graph B) digraph C) multi graph D) simple graph
6. Let A be the adjacency matrix of a simple graph G. Then the ijth entry in A^5 is the number of different edge sequences of _____ edge between vertices v_i and v_j .

   A) 6 B) 5 C) 3 D) 4
7. Distinct left cosets of H={0,3} of group (Z_6, $+_6$) is

   A) 1+H , 2+H, 3+H B) 0+H , 1+H, 2+H

   C) 2+H , 3+H, 4+H D) 0+H , 2+H, 3+H
8. Give an example of a semigroup which is not a monoid.
9. For every 'a' in the lattice (A, ≤) , a ∨ a = ____ and a ∧ a = ____ .
10. Let X = $\{2,3,6,12,24,36\}$ and the relation ≤ be such that $x \le y$, if x divides y. Draw the Hasse diagram of (X, ≤).

PART B – (5 × 16 = 80 Marks)

11. a) i. Let $X = \{1, 2, 3, 4, 5, 6, 7\}$ and $R = \{(x, y) \ / \ (x - y) \text{ is divisible by } 3\}$. Show that R is an equivalence relation. (8 Marks)

11. a) ii. Let the functions 'f' and 'g' be defined as $f(x) = 2x + 1$ and $g(x) = x^2 - 2$. Determine the composition function $f \circ f$, $g \circ g$, $f \circ g$ and $g \circ f$. (8 Marks)

OR

11. b) i. In a survey of 100 students, it was found that 40 studied Mathematics, 64 studied Physics, 35 studied Chemistry, 1 studied all the three subjects, 25 studied Mathematics and Physics, 3 studied Mathematics and Chemistry and 20 studied Physics and chemistry. Find the number of students who studied only Chemistry and the number of students who studied none of these subjects. (8 Marks)

11. b) ii. Let $A = \{1, 2, 3\}$ and R, S be defined on A such that

$$M_R = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad M_S = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

Find the matrices of

A) $(R \cup S)^C$ B) $(R \cap S)^{-1}$ C) $(R \circ S)^{-1}$ D) R^2 E) $R \oplus S$ (8 Marks)

12. a) i. Show that $p \vee (q \wedge r)$ *and* $(p \vee q) \wedge (p \vee r)$ are logically equivalent. (8 Marks)

12. a) ii. Show that $(\exists x)(F(x) \to \neg S(x))$ follows from the premises

$(\exists x)(F(x) \wedge S(x)) \to (y)(M(y) \to W(y)), (\exists y)(M(y) \wedge \neg W(y))$ (8 Marks)

OR

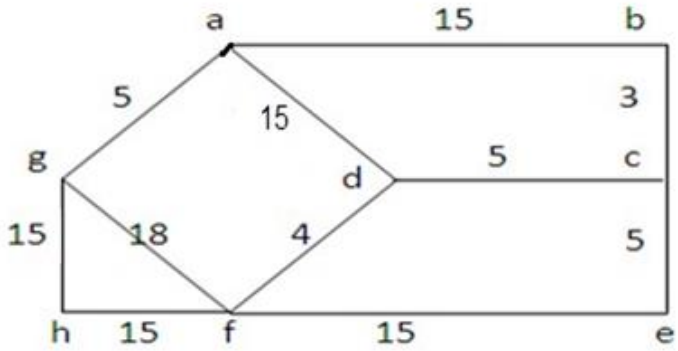
12. b) i. Prove that the following argument is valid.

$p \to \neg q, r \to q, r \Rightarrow \neg p$. (8 Marks)

12. b) ii. Show that the premises " One student in this class knows how to write programs in JAVA" and " Everyone who knows how to write programs in JAVA can get a high paying job" imply the conclusion " Someone in this class can get a high paying job". (8 Marks)

13. a) i. Prove that a simple graph with 'n' vertices and 'k' components can have at most $\frac{(n-k)\ (n-k+1)}{2}$ edges. (8 Marks)

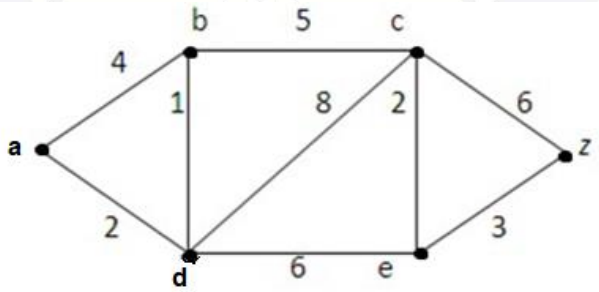
13. a) ii. Using Kruskal's algorithm, find the minimal spanning tree for the weighted graph given below and also find the cost of it.



(8 Marks)

OR

13. b) i. Use Dijkstra's algorithm to find the length of a shortest path between *a* and *z* in the weighted graph given below



(8 Marks)

13. b) ii. Prove that any connected graph with 'n' vertices and 'n – 1' edges is a tree. (8 Marks)

14. a) i. If * is the binary operation on the set R of real numbers defined by a * b = a + b + 2ab, show that (R, *) is a commutative semigroup. (8 Marks)

14. a) ii. State and prove Lagrange's theorem. (8 Marks)

OR

14. b) i. If $f:(G,*)\to(G',\circ)$ is a group homomorphism, then prove that kernel of f is a normal subgroup of G. (8 Marks)

14. b) ii. Prove that the set $Z_4=\{0,1,2,3\}$ is a commutative ring with respect to the binary operation $+_4$ and $\times_4$. (8 Marks)

15. a) i. Show that in a lattice if $a\le b\le c$ then,

A) $a\oplus b=b*c$

B) $(a*b)\oplus(b*c)=b=(a\oplus b)*(a\oplus c)$ (8 Marks)

15. a) ii. Show that in a distributive and complemented lattice, $a\le b\Leftrightarrow a*b'=0\Leftrightarrow a'\oplus b=1\Leftrightarrow b'\le a$ (8 Marks)

OR

15. b) i. Draw the Hasse diagram representing the partial ordering $\{(A,B):A\subseteq B\}$ on the power set P(S) where $S=\{a,b,c\}$. Find the maximal, minimal, greatest and least elements of the poset. (8 Marks)

15. b) ii. In a Boolean algebra, show that

$(a+b')(b+c')(c+a')=(a'+b)(b'+c)(c'+a)$. (8 Marks)